

C=0

P1

A (0, P1)

B (1, P1)

C (2, P1)

D (3, P1)

(1, P1)

C=0

P2

E (0, P2)

F (1, P2)

G (2, P2)

H ($\max(1, 3) + 1$, P2)

I (5, P2)

→ K (6, P2)

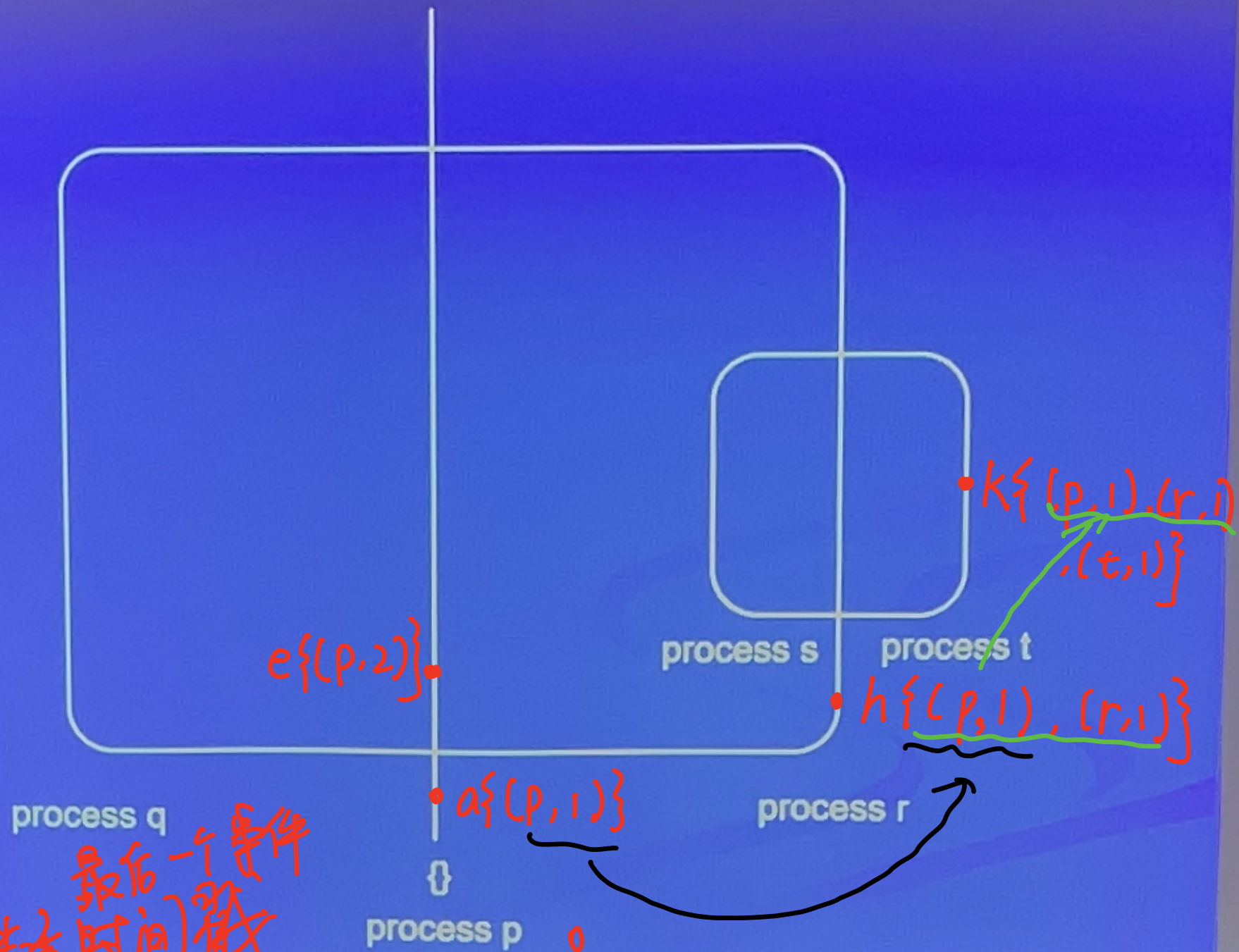
P2 crash
check.

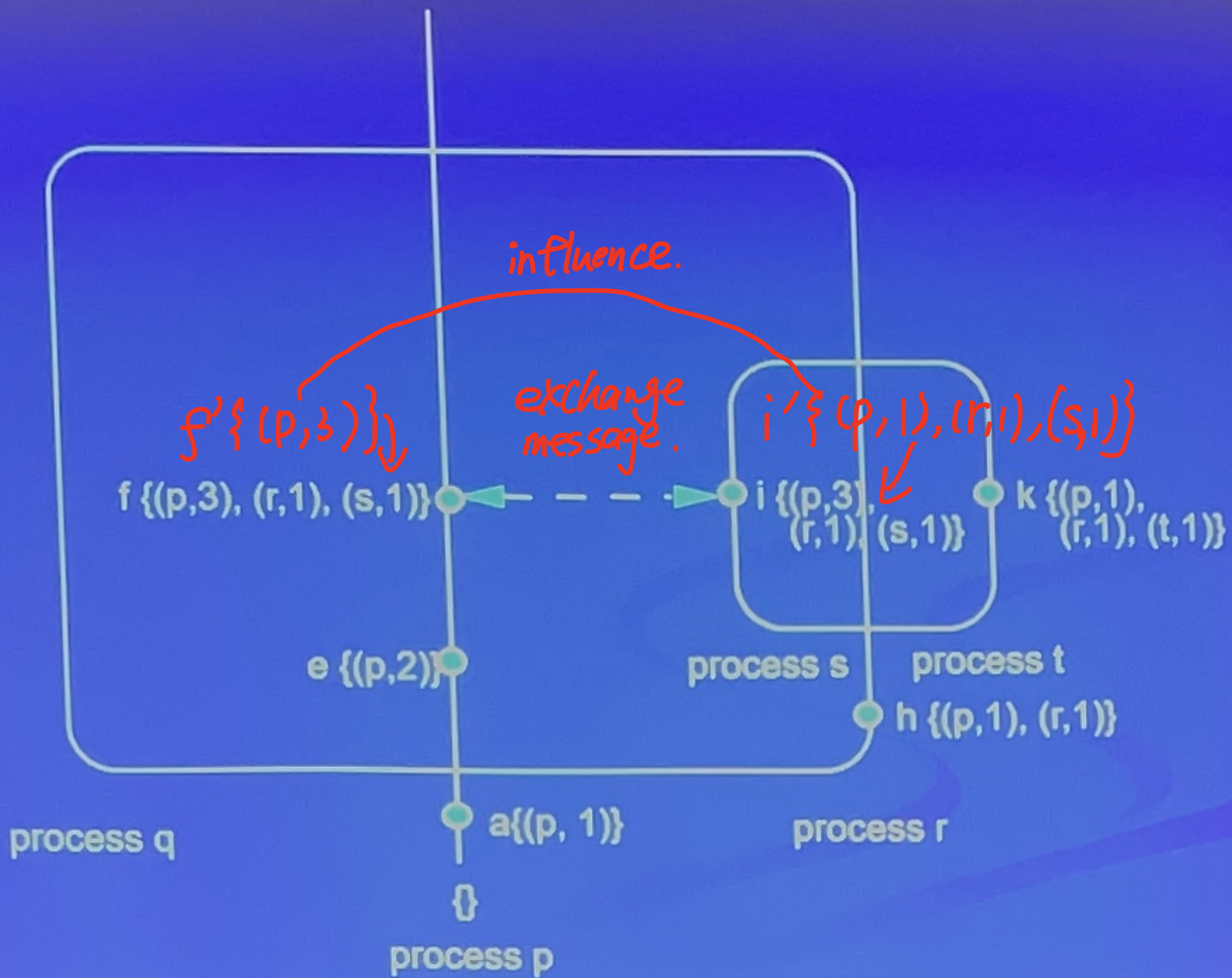


A, B, E ~ ... K

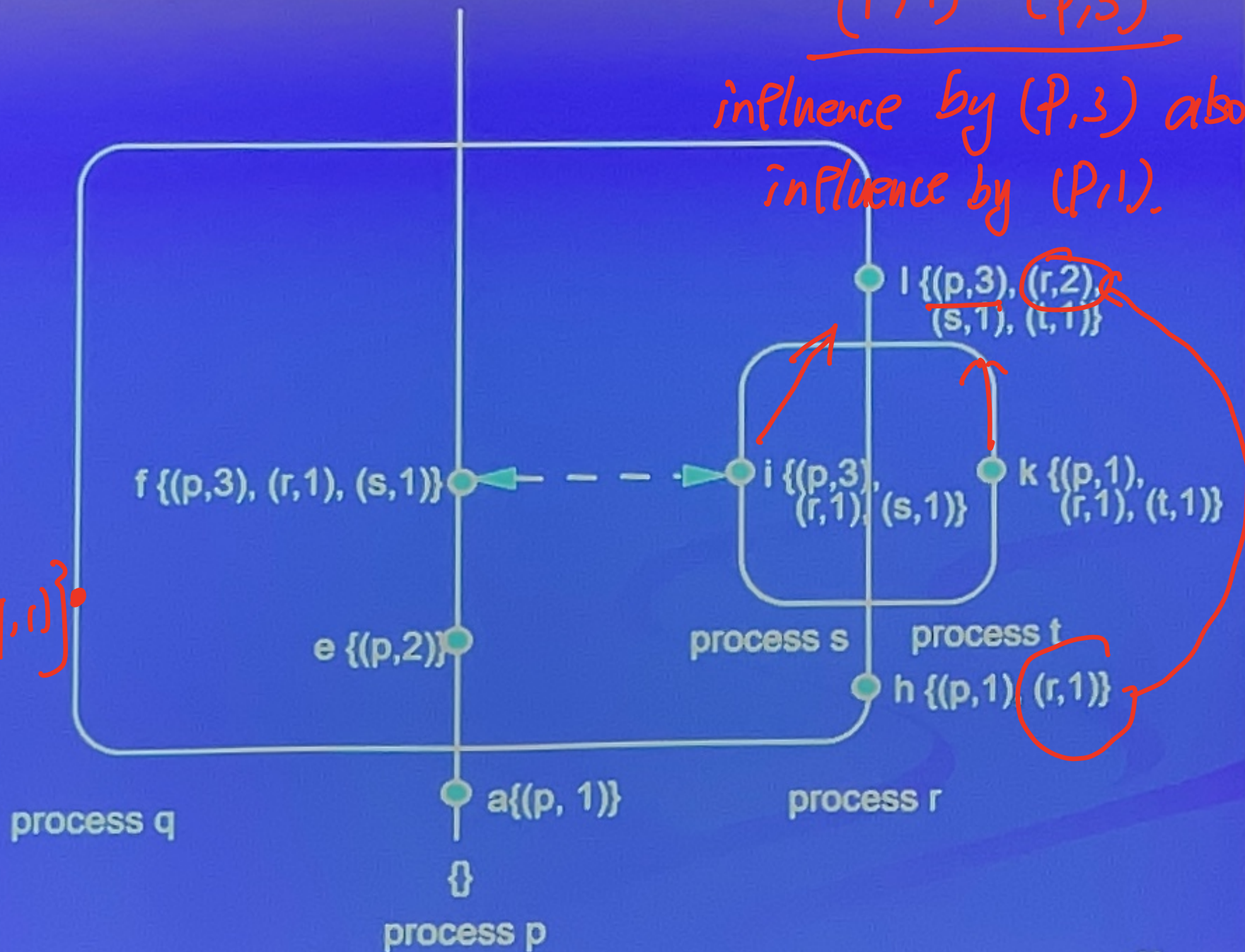
Partially ordered logical clock

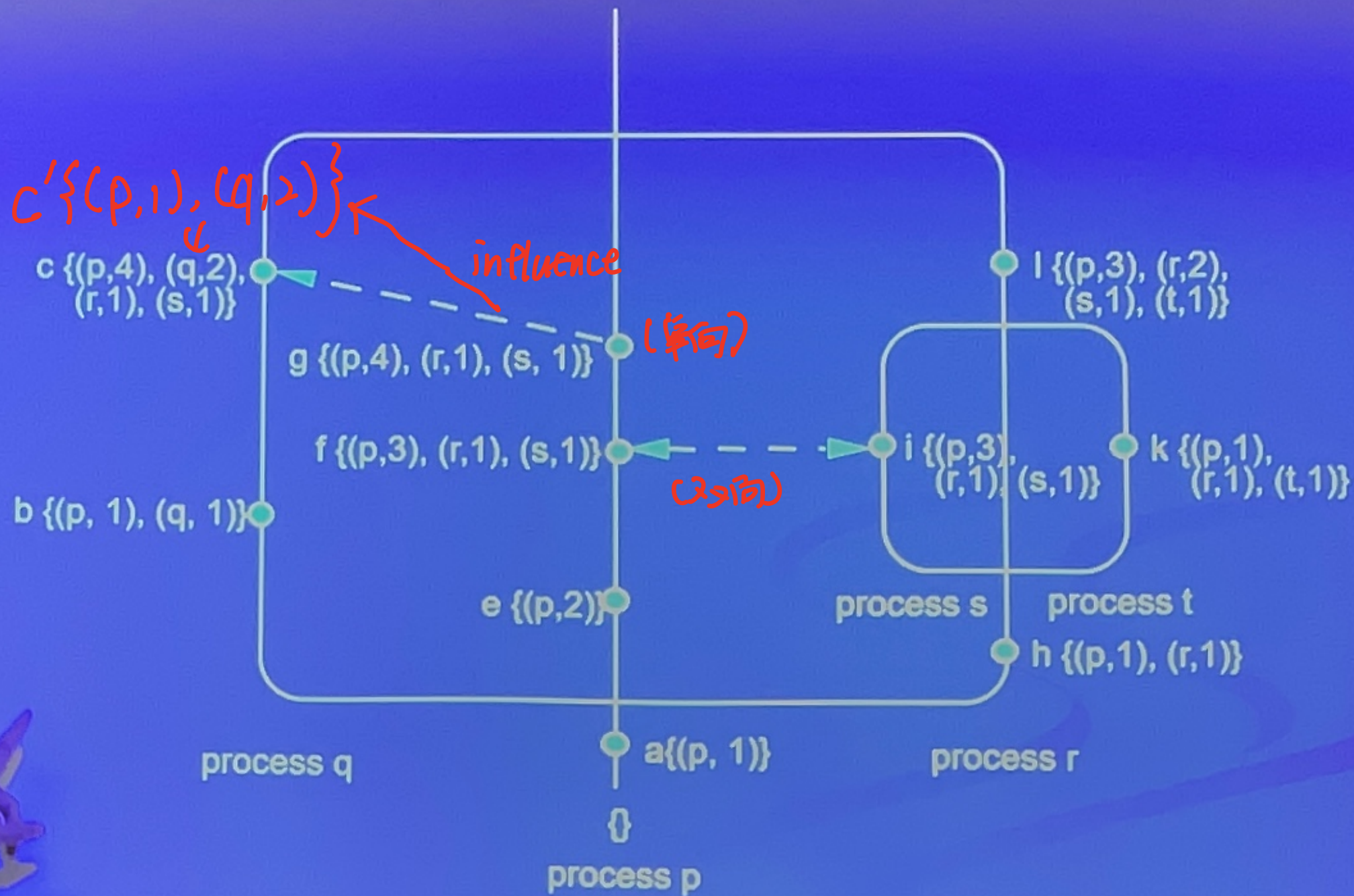
- For some applications, giving an order between each event is **neither necessary nor desirable**.
 - Does not express the causal relations of the events in the system.
 - Debugging a distributed program.
- Partially ordered logical clock allows us to find the causal relation among the processes.
 - The timestamp of the events in a system which uses the partially ordered logical clock is a set of timestamps.
 - Each element in the set is a timestamp of the process which has causal relation with the process.

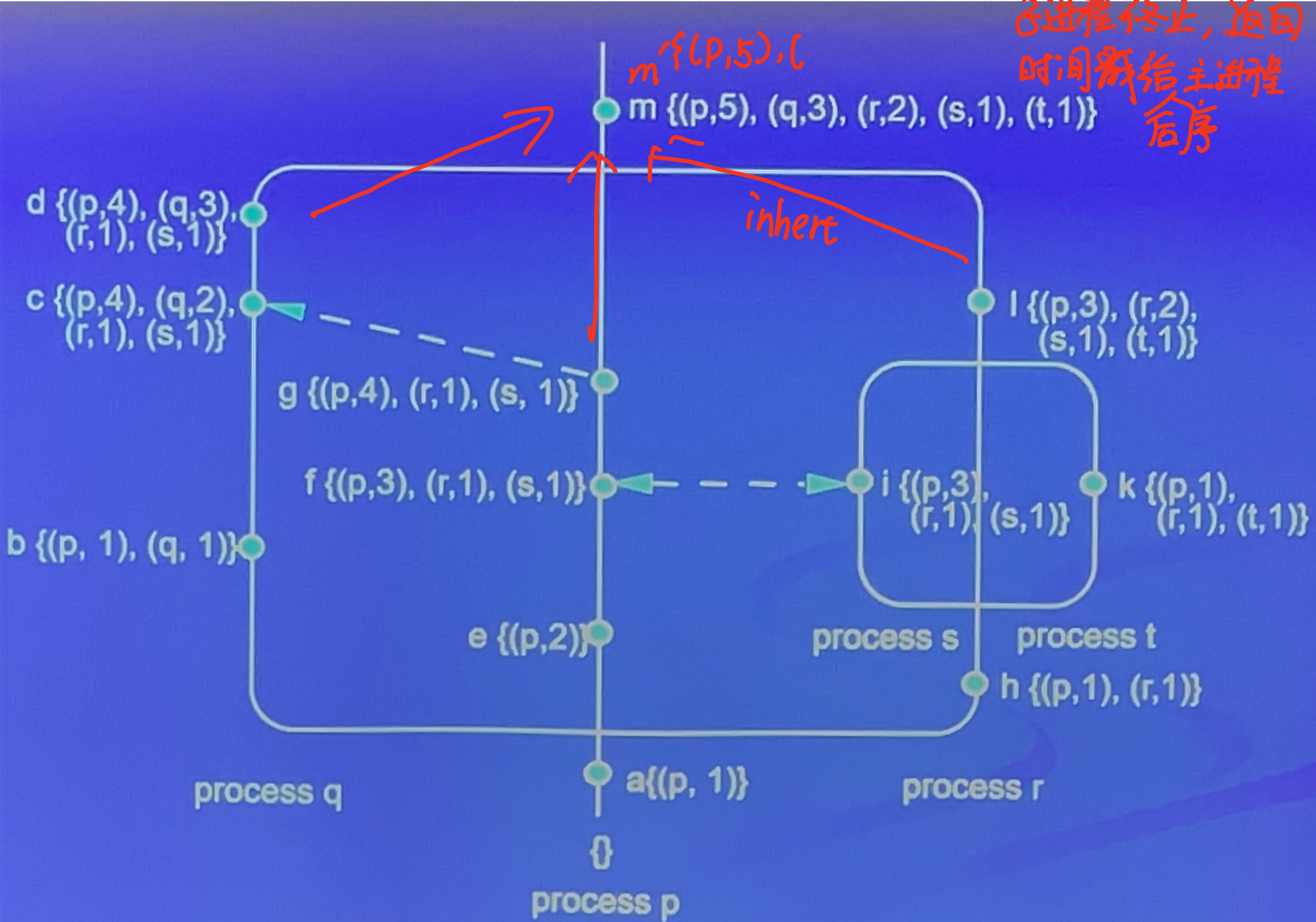




$b\{(p,1), (q,1)\}$







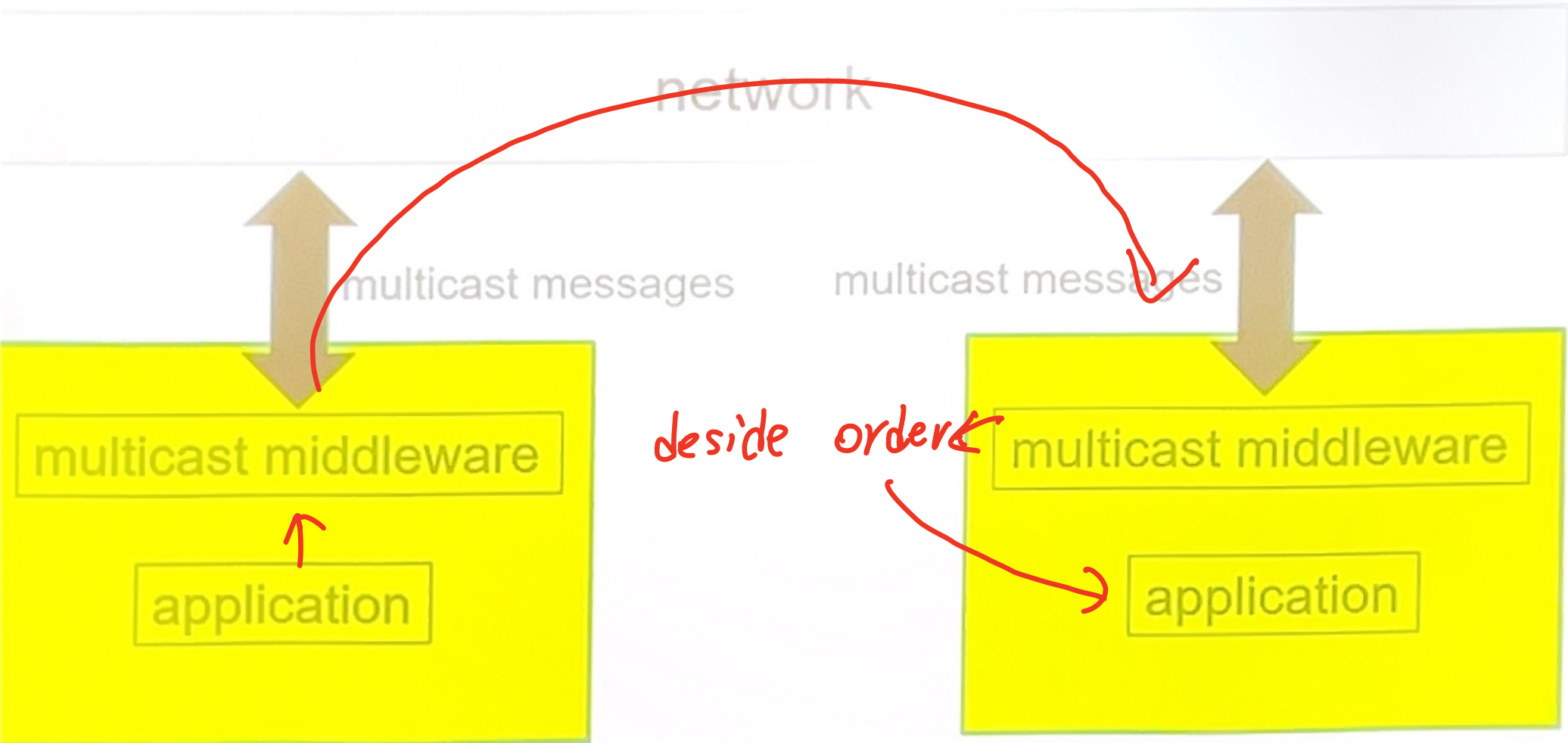
depends on APP.

- Compared with totally ordered logical clock, partially ordered logical clock is more expressive.
- Partially ordered logical clocks may be impractical and expensive for long-live computations.
 - There is no upper bound on the size of the timestamp.
 - The number of item in the timestamp set is only limited by the number of processes created at run time.

Multicast 多点传送

- ❑ A **process group** consists of several processes
 - A process might belong to **multiple groups**
- ❑ A **multicast** operation sends a message to all the members in a group.
- ❑ The **aim** of a multicast operations is, for each process in a group, the process should receive a copy of the message sent to the group.
 - Various delivery guarantees might be required.

The meaning of deliver



Ordered multicast

- **FIFO ordering**: if a process multicasts m and then multicasts m' , then all processes in the group that delivers m' will deliver m before m'